Yuchen Hou

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Education Background

2 degrees

1 National University of Singapore — **Singapore**
Master of Science in Electrical Engineering — *Aug 2025 – Present*

- **Research Experience:** Core member of the **NUS CORE Lab (COntrol, Robotics & Artificial IntElligence)**, focusing on **Embodied AI** and autonomous robotic decision-making.
- **Research Focus:** Perception and planning for aerial robots based on **Vision-Language Navigation (VLN)**, and the application of **End-to-end Reinforcement Learning** in complex 3D environments.

2 Nanjing University of Aeronautics and Astronautics — **Nanjing, China**
Bachelor of Engineering in Electronic and Communication Engineering — *Sep 2021 – Jul 2025*

- **GPA:** 88/100
- **Key Courses (with Scores):** Advanced Mathematics (96), Probability Theory and Mathematical Statistics (97), Random Signal Analysis (97), Microwave Technology and Antenna (97), Fundamentals of Communication (95), Complex Function (99).

Internship Experience

3 positions

1 Honor Device Co., Ltd. — **Shanghai, China**
VLA Algorithm Engineer (Intern) — *Jun 1 – Aug 31, 2026*

Owned a human-in-the-loop real-robot learning stack for VLA models on Honor's self-developed **Omega humanoid**, end to end across six parts:

① **End-to-end VLA pipeline:** Set up the lab's collect → post-train → deploy loop from scratch (two teleoperation rigs, operator training, **GR00T N1.7** post-training, chunked real-time inference).

② **Offline data design:** Designed four demonstration categories (standard, pick and place perturbations, **failure recovery**) so the policy re-tracks moving targets and recovers from missed grasps.

③ **Human-takeover training infra:** Built **smooth teleoperator takeover**, an **intervention-driven data pipeline** and a distributed **Actor–Learner–Robot** loop (AGX/NX, x86 Actor, GPU Learner); a mirrored-layout task rose from **near 0% to 50%** autonomous success in about 140 episodes.

④ **Real-robot RL algorithms:** Ported **RECAP** ($\pi_{0.6}$) advantage-conditioned training to GR00T; designed **RAPID**, a DSRL-style noise-space adaptation that reuses human corrections before a new checkpoint lands; extended the stack to **loco-manipulation**.

⑤ **ROVE reproduction:** Reproduced XPENG's **ROVE** on fine-grained phone grasping — stage-aware returns plus an **optimistic (expectile) value function** that pins negative advantage on takeover-causing actions; autonomous success from **40% to 52%**.

⑥ **Real-robot debugging:** Traced six system-level bugs (colour-space mismatch, classifier hysteresis, yaw drift, stale camera frames) with quantitative probes.

Result: an in-house loop in which the robot learns new tasks from human takeovers — spanning data, systems, RL algorithms and deployment.

2 Huawei Norbert Wiener Research Center — **Singapore**
Research Intern — *Mar 2026 – May 2026*

- Developed backend **deep learning** models for **short-range radar** signal processing.
- Conducted research on downstream tasks including **human pose estimation** and **indoor mapping/SLAM** from radar data.

3 Movel.AI — **Singapore**
Robotics Software Intern — *Dec 2025 – Jan 2026*

- On-site validation and **factory deployment** of autonomous navigation for **forklift robots** in real warehouse environments (**ROS2**).

Research Experience

4 projects · 2 papers accepted to IROS 2026

1 LangGap: Diagnosing and Closing the Language Gap in Vision-Language-Action Models
First Author · NUS CORE Lab *Oct 2025 – Jun 2026*

- *Status:* **Accepted to IROS 2026**.
- *Preprint:* arXiv:2603.00592
- Discovered that state-of-the-art VLA models achieve high benchmark scores but **ignore language instructions**, revealing a gap between performance and true understanding.
- Designed systematic **semantic perturbation** evaluation framework to expose language grounding failures.
- Proposed **multi-task same-scene training** approach and constructed augmented dataset for fine-tuning.

2 Zero-Shot Indoor Object Navigation for Autonomous Drones Using Dual-Policy Reinforcement Learning
Team Member · NUS CORE Lab *Jun 2025 – Jun 2026*

- *Status:* **Accepted to IROS 2026**.
- *Preprint:* arXiv:2601.15614
- Developed a dual-mode visual navigation system based on **end-to-end Reinforcement Learning**, utilizing **RGB-D sensor** input.
- Engineered the policy to autonomously switch between **'Exploration'** and **'Target-driven Navigation'** modes based on target visibility.
- Trained the agent from scratch to master a **discrete action space**, including complex maneuvers like take-off, navigation, and landing.

3 Vision-Language Navigation on Autonomous Drone
Team Member · NUS CORE Lab *Jun 2025 – Feb 2026*

- *Status:* **Nearing completion**.
- Built a robust pipeline to generate various **3D paths** in the **Habitat simulator**.
- Overcame the challenges of the simulator initially designed only for ground robots by designing a robust **3D navigation algorithm** and obstacle detection method.
- Trained a strong and general policy for **drone navigation**.

4 Frontier Based Autonomous Exploration Vehicle Design with ROS2 and Lidar
Team Leader · NUAA *Sep 2024 – Apr 2025*

- Led a team of two members to develop an autonomous exploration system using **ROS2** and **LiDAR** technology.
- Implemented **SLAM** algorithms including **Cartographer** and **Navigation2**, achieving real-time mapping and path planning.
- Integrated **YOLOv11** for object detection.
- Successfully deployed the complete system on an **embedded hardware platform**.

Skills & Awards

Applied Skills

- **Programming & Systems:** **Python** (Proficient), **Linux (Ubuntu)**, Shell scripting.
- **AI Frameworks & Simulation:** **PyTorch** (DRL policies), **Habitat**, **AI2-THOR** (Complex 3D environments).
- **Robotics Development:** **ROS2** (Nodes, Comm), **SLAM** (Cartographer, Nav2), **YOLO** integration.

Research Capabilities

- **Academic & Theory:** **Sim-to-Real Transfer**, Embodied AI Architectures, **RL Algorithms**.
- **Literature & Writing:** Tracking top-tier conferences (**CVPR/IROS/ICRA**), Academic Paper Writing.

Languages

- **English (Proficient):** IELTS Overall 7.0 (Reading: 8.5, Writing: 7.5, Listening: 6.5, Speaking: 6.0).
- **Mandarin:** Native Speaker.

Awards

- Second-class Scholarship (2024) and Third-class Scholarship (2021, 2022, 2023) at Nanjing University of Aeronautics and Astronautics (NUAA).